Supporting Information

Intermittent Micro-Aeration Enhanced Anaerobic Digestion: The Key Role of Fe(III)/Fe(II) Cycle and Reactive Oxygen Species

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This supporting information contains 19 pages, including 5 texts, 5 tables, and 4 figures.

Text S1 The selection of ferrihydrite dosing and the intensity/frequency of micro-aeration

According to our previous study, a low Fe(III) content (< 750 mg/L) promoted anaerobic digestion, while as the Fe(III) content increased to 1470 mg/L, methane production was slightly inhibited (Wang et al. 2024). Therefore, 15 mM Fe(III) in the form of ferrihydrite, containing 840 mg/L Fe(III), was selected here.

To prevent the potential inhibition of methanogens by excessive aeration, this study adopted a low aeration intensity (14 mL of O_2/($L_{Reactor}$·day)). Furthermore, different aeration strategies have been reported; air or oxygen can be dosed either one time, intermittently (pulse-mode), or continuously (Botheju and Bakke 2011). To achieve multiple redox alternations for Fe(II) oxidation and ROS generation, the anaerobic system was aerated for 30 min every 3 h.

Text S2 ROS measurement

The •OH formation pathways were explored by examining the intermediates (e.g., $•O_2^-$ and H_2O_2). •OH and $•O_2^-$ production was qualitatively detected by an electron paramagnetic resonance (EPR) spectrometer with DMPO as a spin-trapping agent. The formation of •OH and H_2O_2 was determined using a fluorescence probe method with 50 μM coumarin and 200 μM cerium sulfate (Liu et al. 2023, Wu et al. 2023).

The parameters of electron paramagnetic resonance (EPR) spectrometer

EPR instrument (Bruker, Elexsys II E500-CW-EPR) operating parameters were center field 3500 G; microwave frequency 9.8 GHz; microwave power 2.0 mW; modulation frequency 4.0 G; modulation amplitude 4.0 G; sweep width 200 G; receiver gain 3.54×10^4; time constant 41.0 ms; sweep time 15 s; 15 scans.

Quantitative analysis of H_2O_2 and •OH

The H_2O_2 concentration was measured using the cerium sulfate titration method, where the yellow Ce^{4+} solution would be reduced by H_2O_2 to colorless Ce^{3+} following the stoichiometry of $2Ce^{4+} + H_2O_2 = 2Ce^{4+} + 2H^+ + O_2$. The concentration of Ce^{4+} before and after the redox reaction can be detected by ultraviolet-visible spectroscopy. The test value at 316 nm was adopted. 1 mM of Ce^{4+} solution was prepared by dissolving 332 mg of $Ce(SO_4)_2$ in 0.5 M of H_2SO_4 solution(1L). The H_2O_2 concentration is calculated by the equation of

$$C(H_2O_2)= (C_0(Ce^{4+})\times V(Ce^{4+}) - C(Ce^{4+})\times(V(Ce^{4+})+V(H_2O_2)))/(2\times V(H_2O_2))$$

The oxidation product of coumarin by •OH, i.e., 7-hydroxylcoumarin (7h-COU), was measured at an excitation of 350 nm and an emission of 460 nm using a microplate reader. Then, the concentration of •OH was calculated according to 7h-COU/14.5%.

Text S3 Characterization and analysis

1. Morphology characterization

X-ray diffraction (XRD) was applied for the identification of the crystalline iron phases in digesters. The XRD diffractometer (SmartLab 9kw, Japan) equipped a Cu-Ka radiation (45 kV, 200 mA) at a scan rate was 8°/min, between 5° and 80° (Wang et al. 2022).

2. Electrochemical measurement

After washing three times with 0.01 M phosphate buffer solution (PBS), 20 μL of the sediment suspensions were coated on a glassy carbon electrode (Φ3 mm, working area: 0.07 mm2) and dried in an infrared oven. Electrochemical tests were performed with an electrochemical workstation (Chenhua, CHI 660). The coated glassy carbon electrode, Ag/AgCl electrode, and Pt electrode were selected as the working electrode, reference electrode, and counter electrode, respectively. The capacitance and resistance of the sediment were assessed with cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS), respectively.

Cyclic voltammetry was performed in the range of -0.8 to 0.8 V (vs Ag/AgCl) at scan rates of 50 mV/s, with 30 mL of 0.01 M PBS as the electrolyte. Electrochemical impedance spectroscopy was conducted over a frequency range of 1000 kHz to 0.01 Hz, with a sinusoidal perturbation of 10 mV amplitude to analyze the internal impedances of the sediments. Before the electrochemical tests, the electrolyte in the same concentration was aerated for 40 min with N2 to maintain an anaerobic condition.

3. Analytical methods

TSS, VSS, and COD were determined by the Standard Methods for the Examination of Water and Wastewater (Wang et al. 2018). The volume and content of biogas were measured using a 100 mL glass syringe and a gas chromatograph (Tianmei,

GC-7900, China) with a thermal conductivity detector (TCD), respectively (Zhao et al. 2015).

Determination of total Fe(II)

A 0.5 mL homogeneous sample was taken from the reactor to detect Fe(II) concentration. 1 mL HCl (6 M) was added to the sample immediately after collection to inhibit the oxidation of Fe^{2+} to Fe^{3+}. Total Fe(II) content was analyzed by a modified 1,10-o-phenanthroline analytical method. Briefly, after two hours of acid leaching, 0.5 mL supernatant after centrifugation was mixed with 1 mL hydroxylamine hydrochloride solution (10% w/v) and boiled for 20 min to quantify total Fe. Then the sample was mixed with 5 mL iron buffer solution (40% w/v ammonium acetate in 50% v/v glacial acetic acid) and 2 mL 1,10-o-phenanthroline solution (0.5% w/w). The 1,10-o-phenanthroline was given 15 min to react with Fe(II) before absorbance at 510 nm was measured.

Determination of phenol

The concentration of phenol was quantified using HPLC equipped with UV–vis detector (Waters, 2489) and a C18 column (5 μm, 4.6 mm × 250 mm), the mobile phase (flow rate of 1 mL/min) was the mixture consisting of methanol and H_2O (70:30, V/V), and determined by UV mode at 282 nm, respectively.

Text S4 Metagenomic sequencing and analysis.

Total DNA from aggregates in each UASB digester was extracted by E.Z.N.A™ Mag-Bind Soil DNA Kit (OMEGA, M5635-02, USA). The concentrations of the extracted DNA were determined by a spectrophotometer (Thermo Fisher, Q33238, USA). Fragmentation buffer solutions were added into the qualified DNA samples that were randomly interrupted into the short DNA fragments by ultrasound. The short DNA fragments were used for library construction. Metagenomic sequencing was conducted by Sangon Biotech (Shanghai) Co. Ltd, China using an Illumina HiSeq 2500 platform. To ensure the reliability of data, the raw reads were pre-processed using a Trimmomatic software to obtain high-quality clean reads (Bolger et al. 2014). Quality control was performed with FastQC (Schmieder and Edwards 2011). Clean reads were assembled using a MEGAHIT software to obtain contigs (Li et al. 2015). Open reading frames (ORFs) in the contigs were predicted using a Prodigal software. Gene clustering and redundancy removal were performed to obtain a non-redundant gene catalogue (Unigenes) using a Mmseqs software (Steinegger and Soeding 2017). Clean reads after quality control were blasted with the gene catalogue to calculate the different Unigene levels using a bbmap software (Holmes et al. 2017). Protein sequences were aligned to KEGG database to obtain the functional information (Huerta-Cepas et al. 2017).

Text S5 Economic Analysis of the Process of Intermittent Micro-aeration Coupled with Ferrihydrite Addition

The economic feasibility was evaluated on a 1 m^3 reactor with the same O_2 supply and ferrihydrite addition as the experiment in this study. In practical engineering applications, O_2 could be replaced by ambient air with equivalent O_2 content to save cost. The cost primarily consists of aeration expenses and ferrihydrite procurement costs, while the revenue can be derived from the increased methane production.

Specifically, the aeration energy consumption was 0.1 CNY/day (50 W × 4 h/day × 0.5 CNY/kW·h). The cost of a single ferrihydrite addition (15 mM Fe(III), calculated as $Fe_5HO_8 \cdot 4H_2O$) was 43.2 CNY (15 mM × 1000 L × 0.096 kg × 30 CNY/kg). The daily revenue generated from methane yield increase amounts to 1.22 CNY/day (0.67 m^3/day × 1.82 CNY/m^3). Therefore, the system reached break-even after approximately 39 days (43.2 CNY /(1.22 CNY/day - 0.1 CNY/day)).

Table S1. Performance parameters of the reactor during operation

Stages	Influent COD, mg/L	Glucose, mg COD/L	Phenol, mg COD/L	COD removal rate, %		Phenol removal rate, %		CH_4 production, mL/day	
				IMA [a]	Control [b]	IMA	Control	IMA	Control
0-10 d	2500	2500	-	69.00%	66.13%	-	-	252.11	255.29
11-32 d	5500	5500	-	71.64%	67.56%	-	-	800.71	742.32
33-40 d	8000	8000	-	90.99%	83.23%	-	-	1860.95	1431.20
41-58 d	8000	6800	1200	86.18%	87.66%	87.18%	86.01%	1565.9	1295.51
59-70 d	8000	5600	2400	90.67%	88.20%	81.70%	81.15%	1828.5	1611.87
71-124 d	8000	3200	4800	82.20%	68.57%	73.53%	48.69%	1602.0	1233.22

[a] IMA: Intermittent micro-aeration group

[b] Control: Control group

Table S2. The typical genes involved in dissimilatory iron reduction and their abundance based on KEEG.

K Number	Description	Abundance IMA[a]	Control[b]
CYC-like proteins			
K03385	nrfA; nitrite reductase (cytochrome c-552) [EC:1.7.2.2]	1.38E-02	8.81E-03
K15876	nrfH; cytochrome c nitrite reductase small subunit	6.03E-03	3.11E-03
K02274	coxA, ctaD; cytochrome c oxidase subunit I [EC:7.1.1.9]	1.85E-03	2.33E-03
K02276	coxC, ctaE; cytochrome c oxidase subunit III [EC:7.1.1.9]	0.00E+00	1.10E-04
K00404	ccoN; cytochrome c oxidase cbb3-type subunit I [EC:7.1.1.9]	1.02E-04	1.91E-04
K00412	CYTB, petB; ubiquinol-cytochrome c reductase cytochrome b subunit	4.69E-04	2.08E-04
K00411	UQCRFS1, RIP1, petA; ubiquinol-cytochrome c reductase iron-sulfur subunit [EC:7.1.1.8]	1.42E-04	7.12E-05
K00405	ccoO; cytochrome c oxidase cbb3-type subunit II	4.46E-05	8.81E-05
mtrCAB			
K00577	mtrA; tetrahydromethanopterin S-methyltransferase subunit A [EC:2.1.1.86]	9.69E-02	7.06E-02
K00578	mtrB; tetrahydromethanopterin S-methyltransferase subunit B [EC:2.1.1.86]	3.47E-02	2.26E-02
K00579	mtrC; tetrahydromethanopterin S-methyltransferase subunit C [EC:2.1.1.86]	4.14E-02	2.80E-02
ribDCBAE			
K14652	ribBA; 3,4-dihydroxy 2-butanone 4-phosphate synthase / GTP cyclohydrolase II [EC:4.1.99.12 3.5.4.25]	6.02E-02	3.48E-02
K11752	ribD; diaminohydroxyphosphoribosylaminopyrimidine deaminase / 5-amino-6-(5-phosphoribosylamino)uracil reductase [EC:3.5.4.26 1.1.1.193]	8.45E-03	5.03E-03
K00793	ribE, RIB5; riboflavin synthase [EC:2.5.1.9]	5.21E-02	3.26E-02

[a] IMA: Intermittent micro-aeration group.

[b] Control: Control group.

[c] Data was represented as mean from three replicate measurements. The red font indicated significant differences between the two groups (p<0.05).

Table S3. The typical genes that encode V/A-/F-type ATPase and their abundance based on KEEG.

Module and Pathway	K Number	Description	Abundance	
			IMA[a]	Control[b]
M00157, F-type ATPase, prokaryotes and chloroplasts	K02111	ATPF1A, atpA; F-type H+/Na+-transporting ATPase subunit alpha	1.29E-01[c]	1.25E-01
	K02112	ATPF1B, atpD; F-type H+/Na+-transporting ATPase subunit beta	1.66E-01	1.56E-01
	K02113	ATPF1D, atpH; F-type H+-transporting ATPase subunit delta	6.23E-02	8.26E-02
	K02114	ATPF1E, atpC; F-type H+-transporting ATPase subunit epsilon	6.24E-02	7.23E-02
	K02115	ATPF1G, atpG; F-type H+-transporting ATPase subunit gamma	7.65E-02	9.80E-02
	K02108	ATPF0A, atpB; F-type H+-transporting ATPase subunit a	8.30E-02	9.64E-02
	K02109	ATPF0B, atpF; F-type H+-transporting ATPase subunit b	7.00E-02	7.66E-02
	K02110	ATPF0C, atpE; F-type H+-transporting ATPase subunit c	6.22E-02	6.30E-02
M00159, V/A-type ATPase, prokaryotes	K02117	ATPVA, ntpA, atpA; V/A-type H+/Na+-transporting ATPase subunit A	8.11E-02	6.44E-02
	K02118	ATPVB, ntpB, atpB; V/A-type H+/Na+-transporting ATPase subunit B	1.21E-01	8.83E-02
	K02119	ATPVC, ntpC, atpC; V/A-type H+/Na+-transporting ATPase subunit C	4.81E-02	3.34E-02
	K02120	ATPVD, ntpD, atpD; V/A-type H+/Na+-transporting ATPase subunit D	6.82E-02	5.11E-02
	K02121	ATPVE, ntpE, atpE; V/A-type H+/Na+-transporting ATPase subunit E	4.57E-02	2.91E-02
	K02122	ATPVF, ntpF, atpF; V/A-type H+/Na+-transporting ATPase subunit F	3.49E-02	2.66E-02
	K02107	ATPVG, ahaH, atpH; V/A-type H+/Na+-transporting ATPase subunit G/H	2.88E-02	1.77E-02
	K02123	ATPVI, ntpI, atpI; V/A-type H+/Na+-transporting ATPase subunit I	5.31E-02	3.60E-02
	K02124	ATPVK, ntpK, atpK; V/A-type H+/Na+-transporting ATPase subunit K	5.09E-02	4.41E-02

[a] IMA: Intermittent micro-aeration group.

[b] Control: Control group.

[c] Data was represented as mean from three replicate measurements. The red font indicated significant differences between the two groups (p<0.05).

Table S4. The typical genes involved in four methanogenic pathways and their abundance based on KEEG.

Module and Pathway	EC Number	K Number	Description	Abundance	
				IMA[a]	Control[b]
M00567, Methanogenesis, CO_2 => methane	1.2.7.12	K00202	fwdC, fmdC; formylmethanofuran dehydrogenase subunit C	1.04E-01	6.61E-02
		K00205	fwdF, fmdF; 4Fe-4S ferredoxin	4.99E-02	3.85E-02
		K00203	fwdD, fmdD; formylmethanofuran dehydrogenase subunit D	7.05E-02	4.67E-02
		K11260	fwdG; 4Fe-4S ferredoxin	2.82E-02	1.81E-02
		K00200	fwdA, fmdA; formylmethanofuran dehydrogenase subunit A	5.18E-02	3.69E-02
		K11261	fwdE, fmdE; formylmethanofuran dehydrogenase subunit E	1.05E-01	6.98E-02
		K00201	fwdB, fmdB; formylmethanofuran dehydrogenase subunit B	1.19E-01	7.27E-02
	2.3.1.101	K00672	ftr; formylmethanofuran--tetrahydromethanopterin N-formyltransferase	4.89E-02	3.71E-02
	3.5.4.27	K01499	mch; methenyltetrahydromethanopterin cyclohydrolase	5.35E-02	4.04E-02
	1.5.98.1	K00319	mtd; methylenetetrahydromethanopterin dehydrogenase	4.09E-02	3.44E-02
	1.5.98.2	K00320	mer; 5,10-methylenetetrahydromethanopterin reductase	1.26E-01	8.14E-02
	1.12.98.1	K00440	frhA; coenzyme F420 hydrogenase subunit alpha	2.12E-02	2.69E-02
		K00442	frhD; coenzyme F420 hydrogenase subunit delta	1.05E-02	1.28E-02
		K00443	frhG; coenzyme F420 hydrogenase subunit gamma	1.82E-02	2.30E-02
		K00441	frhB; coenzyme F420 hydrogenase subunit beta	2.33E-02	2.41E-02
M00357, Methanogenesis, acetate => methane	2.7.2.1	K00925	ackA; acetate kinase	8.49E-02	1.08E-01
	2.3.1.8	K13788	pta; phosphate acetyltransferase	9.44E-03	3.65E-03
		K00625	E2.3.1.8, pta; phosphate acetyltransferase	1.40E-02	1.40E-02
	6.2.1.1	K01895	ACSS1_2, acs; acetyl-CoA synthetase	3.78E-01	2.31E-01
M00356, Methanogenesis, methanol => methane	2.1.1.90	K04480	mtaB; methanol---5-hydroxybenzimidazolylcobamide Co-methyltransferase	3.61E-04	2.98E-04
	-	K14081	mtaC; methanol corrinoid protein	1.70E-04	2.70E-04

M00563, Methanogenesis, methylamine/dimethylamine/trimethylamine => methane	2.1.1.250	K14083	mttB; trimethylamine---corrinoid protein Co-methyltransferase	5.95E-04	5.46E-04
	2.1.1.248	K16176	mtmB; methylamine---corrinoid protein Co-methyltransferase	1.42E-03	1.69E-03

[a] IMA: Intermittent micro-aeration group.

[b] Control: Control group.

[c] Data was represented as mean from three replicate measurements. The red font indicated significant differences between the two groups (p<0.05).

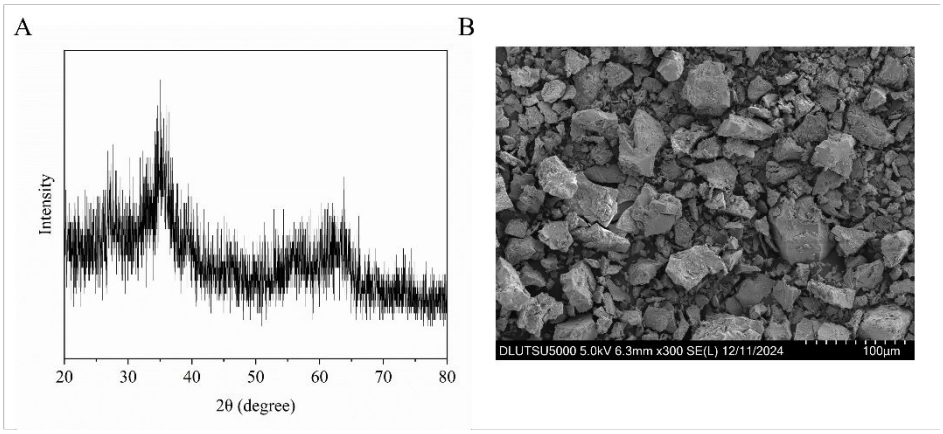
Table S5. The typical genes involved in ROS detoxification and their abundance based on KEEG.

EC Number	K Number	Description	Abundance		Derived from microbes
			IMA[a]	Control[b]	
EC 1.15.1.1	K04564	SOD2; superoxide dismutase, Fe-Mn family	5.88E-02	3.40E-02	*Methanothrix/Methanothrix*
	K04565	SOD1; superoxide dismutase, Cu-Zn family	1.65E-04	1.36E-04	*unclassified_Bacillota_genus/ unclassified_Bacillota_genus*
EC 1.11.1.6	K03781	katE, CAT, catB, srpA; catalase	5.79E-02	7.66E-02	*Methanothrix/ Candidatus_Methanofastidiosum*
EC 1.11.1.24	K13279	PRDX1; peroxiredoxin 1	2.22E-03	5.20E-04	*unclassified_Thermoguttaceae_genus/ unclassified_Thermoguttaceae_genus*
	K03386	PRDX2_4, ahpC; peroxiredoxin 2/4	2.24E-01	1.95E-01	*Methanothrix/Methanothrix*
EC 1.11.1.21	K03782	katG; catalase-peroxidase	1.26E-01	7.74E-02	*unclassified_Myxococcia_genus/ unclassified_Myxococcia_genus*

[a] IMA: Intermittent micro-aeration group.

[b] Control: Control group.

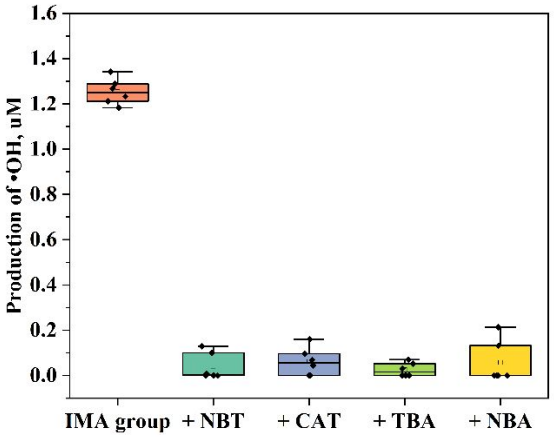
[c] Data was represented as mean from three replicate measurements. The red font indicated significant differences between the two groups (p<0.05).



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170 Fig. S1 (A) XRD spectra of ferrihydrite. (B) The SEM image of ferrihydrite.

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173 Fig. S2 The •OH production in the presence of NBT (Nitro blue tetrazolium), CAT

174 (Catalase), TBA (tert-butanol) and NBA (n-butanol).

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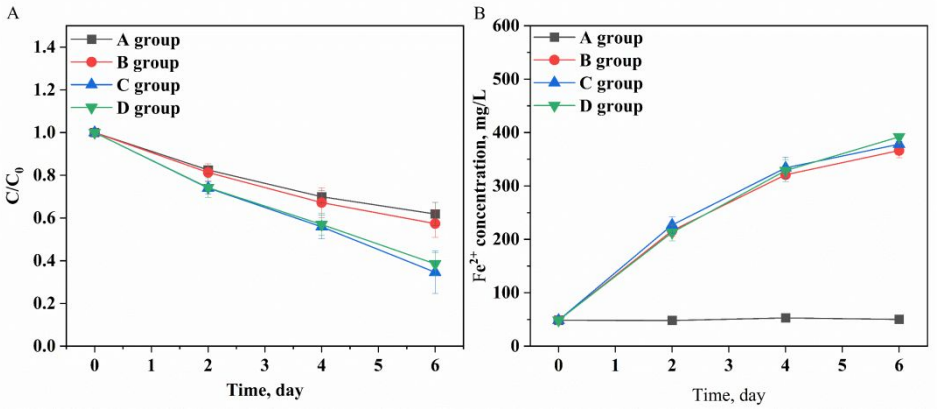
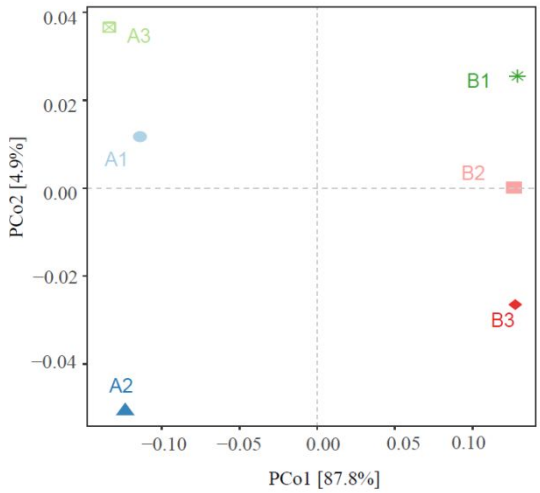


Fig. S3 (A) Degradation of phenol. (B) Generation of Fe^{2+}. Group A: without ferrihydrite addition and micro-aeration, Group B: with ferrihydrite addition, Group C: with ferrihydrite addition and micro-aeration, Group D: with ferrihydrite addition and micro-aeration, and formate (5 mM) as •OH quencher.



181 Fig. S4 Principal co-ordinates analysis (PCoA) analysis with Jaccard distance matrix.

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